

Technical Design Document

Smart Solar Nanogrid Road Lighting System

Virtual Prototype and Proposed System Architecture

1. Project Overview

The proposed system provides energy-efficient road lighting using solar energy, battery storage and intelligent lighting control. An ESP32 processes ambient-light and motion information to control LED brightness.

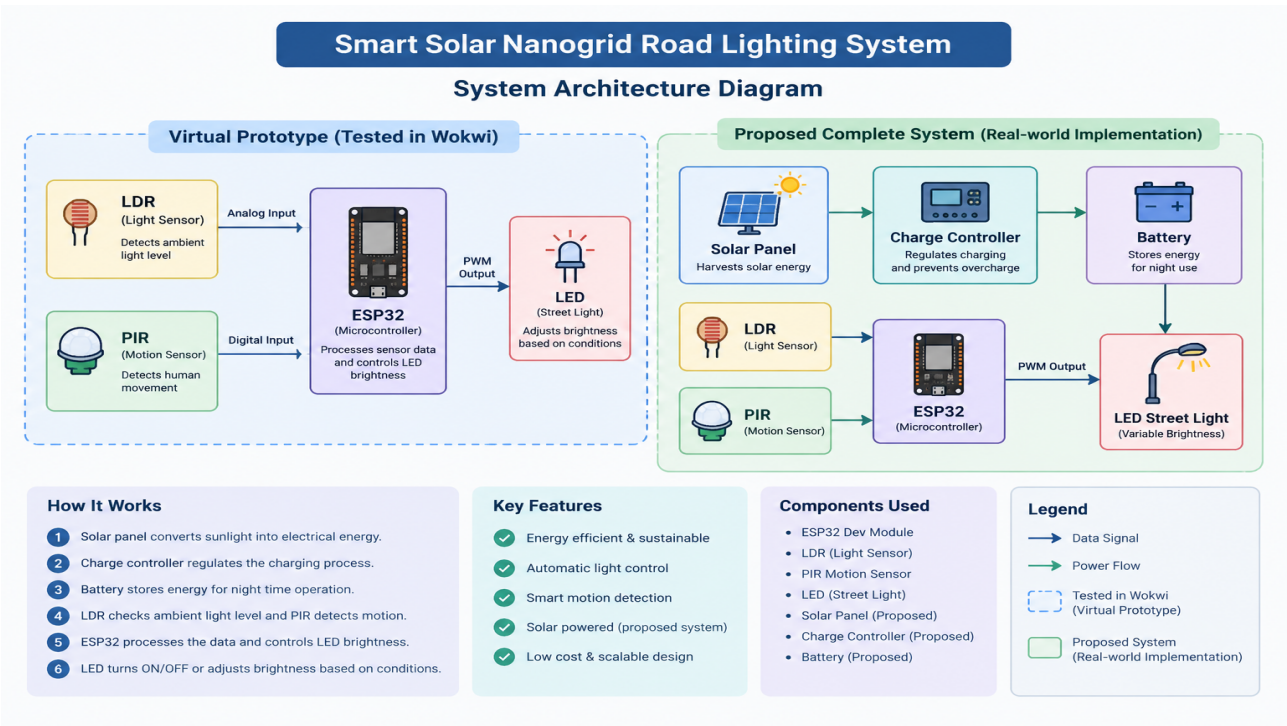
2. Problem Statement

Conventional road lights can waste electricity when operated at full brightness during low-traffic periods. Grid dependence can also reduce reliability in rural and remote areas.

3. Proposed Solution

The system combines a solar power subsystem with smart lighting control. An LDR detects ambient light and a PIR detects movement. During the day the light is OFF; at night it stays dim without movement and becomes bright when movement is detected.

4. System Architecture



Important: The ESP32 + LDR + PIR + LED section is the part actually implemented and tested in Wokwi. Solar panel, charge controller and battery are shown as the proposed complete real-world system and were not physically tested in this prototype.

5. Components

Component	Purpose
ESP32	Reads sensors and controls LED brightness.
LDR / Photoresistor	Measures ambient light.
PIR Motion Sensor	Detects movement.

LED	Represents the street light in Wokwi.
220 Ω Resistor	Limits LED current.
Solar Panel (proposed)	Generates solar energy.
Charge Controller (proposed)	Regulates battery charging.
Battery (proposed)	Stores energy for nighttime operation.

6. Wokwi Pin Configuration

Device	Signal	ESP32
LDR	AO	GPIO 34
PIR	D	GPIO 27
LED	Control	GPIO 25
LDR/PIR	VCC	3.3 V
LDR/PIR/LED	GND	GND

7. Working Principle

- LDR continuously measures ambient light.
- PIR provides a digital signal when movement is detected.
- ESP32 reads both sensors and decides the lighting state.
- Day/high light → LED OFF.
- Night/low light + no motion → LED low brightness.
- Night/low light + motion → LED high brightness.

8. Control Algorithm

Read LDR → Read PIR → Check light level → Check motion → Set LED brightness → Repeat.

9. Software

The prototype uses Arduino/C++ for ESP32 and is simulated in Wokwi. `analogRead()` reads the LDR, `digitalRead()` reads the PIR, and PWM-style LED control provides different brightness levels. Sensor readings are displayed in the Serial Monitor.

10. Prototype Testing

Test	Condition	Expected	Observed	Result
TC01	High light / day	LED OFF	LED OFF	PASS
TC02	Low light + no motion	Low brightness	Low brightness	PASS
TC03	Low light + motion	High brightness	High brightness	PASS

All three intended lighting states were successfully demonstrated in the Wokwi simulation. LDR readings changed with simulated light level, and PIR motion changed the motion state and increased LED brightness under dark conditions.

11. Repository and Reproducibility

The project repository contains the README, ESP32 source code, test cases, license and architecture diagram. The Wokwi simulation link is included in the README for reproducibility.

12. Limitations

The demonstrated prototype is virtual. Physical solar hardware, battery storage, outdoor LED fixtures, weatherproofing and field deployment have not yet been tested.

13. Future Scope

- Build and test the physical solar-powered prototype.
- Add battery state-of-charge and energy monitoring.
- Add fault detection and data logging.
- Connect multiple lighting nodes.
- Add optional IoT-based monitoring.
- Evaluate performance in real outdoor conditions.

14. Conclusion

The virtual prototype demonstrates smart lighting based on ambient light and motion. The proposed solar nanogrid architecture extends this approach toward renewable, reliable and energy-efficient road lighting for rural and remote applications.